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"Can we investigate morphological processing over the web? A web-based replication of a masked morphological priming experiment with non-native speakers"

Bachelor Thesis

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Table of Contents

Zusammenfassung 1

Abstract 2

1. Introduction 3

2. Theoretical background 5

3. Hypotheses 9

4. Study design 10

4.1 Materials 10

4.2 Brief summary of the experimental procedure in Ciaccio and Veríssimo (2020) 11

4.3 Procedure of the online experiment 12

4.4 Participants 14

5. Challenges 15

6. Results of the online study 17

7. Discussion 21

7.1 Prior findings for the masked morphological priming paradigm 21

7.2 Comparison of the results of the present online study and the laboratory study by Ciaccio and Veríssimo (2020) 22

7.3 Limitations 28

8. Conclusion and Outlook 31

9. References 33

10. List of figures and tables 36

Acknowledgements 37

Statement of Originality 38

Appendix 39

Zusammenfassung

Die vorliegende Studie untersucht die Verarbeitung von morphologisch komplexen Wörtern bei Nicht-Muttersprachlern. Sie testet das maskierte morphologische Priming-Paradigma für derivierte und flektierte englische Wörter über das Internet. Um zu prüfen, ob das maskierte morphologische Priming-Paradigma über das Internet untersucht werden kann, wurde in der Online-Studie versucht, die Ergebnisse einer Laborstudie von Ciaccio und Veríssimo (2020) zu replizieren. Die Reaktionszeiten der Online-Studie und der im Labor durchgeführten Studie wurden anhand des Mittelwerts, des Medians und der Standardabweichung verglichen. Diese Werte ermöglichen eine Beurteilung, ob die Reaktionszeiten der Teilnehmer vergleichbar sind. Die Ergebnisse zeigen, dass in beiden Studien Priming-Effekte sowohl für derivierte als auch für flektierte Wörter in nahezu gleichem Ausmaß festgestellt wurden. Obwohl die Reaktionszeiten in der Online-Studie im Allgemeinen länger waren, ähneln die beobachteten Priming-Effekte denen der Laborstudie. Die gefundenen Priming-Effekte stimmen mit vorherigen Studien überein, die sich mit dem Priming-Effekt von derivierten und flektierten Wörtern bei Nicht-Muttersprachlern befassten.

Die Ergebnisse der Online-Studie zeigen, dass das maskierte morphologische Priming-Paradigma bei Nicht-Muttersprachlern über das Internet untersucht werden kann. Dies eröffnet neue Wege für die psycholinguistische Forschung, da mithilfe von Online-Studien mehr Sprecher einer Sprache erreicht werden können und die Möglichkeit besteht, Gruppen von Probanden zu haben, die einen diversen Hintergrund haben, aus unterschiedlichen Regionen kommen und repräsentativer für die Population der Sprecher einer Sprache sind.

Schlüsselwörter: Psycholinguistik, webbasierte Studie, maskiertes morphologisches Priming, Derivation, Flexion

Abstract

The present study investigates the processing of morphologically complex words in non-native speakers. It tests the masked morphological priming paradigm for English derived and inflected words via the web. To examine whether the masked morphological priming paradigm can be investigated over the web, the online study attempted to replicate the results obtained in a laboratory study done by Ciaccio and Veríssimo (2020). The reaction times of the online study and the laboratory study were compared using the mean, median and standard deviation which allow for the assessment whether the reaction times of the participants are comparable. The results show that both studies found priming effects for derived as well as inflected words to an almost similar extent. Although the reaction times in the online study were longer in general, the observed priming effects resemble the priming effects found in the laboratory study. The priming effects that were found align with several previous studies regarding the priming effect of derived and inflected words in non-native speakers.

The results of this online study indicate that the masked morphological priming paradigm with non-native speakers can be investigated over the web. This opens up new avenues for psycholinguistic research since with the help of online studies more speakers of a language can be reached, offering the possibility of having groups of subjects that have different backgrounds, come from different regions and are more representative of the population of speakers of a language.

Keywords: psycholinguistics**,** web-based study, masked morphological priming, derivation, inflection

1. Introduction

Participating in a psycholinguistic study normally involves going to a university or laboratory, meeting a researcher that provides the instructions for an experiment and completing a computer task while sitting in a quiet room or cubicle. During the worldwide Covid-19 pandemic and the following national lockdowns, laboratory work (including psycholinguistic studies) was mostly stopped due to the shutdown of laboratories and universities. Almost all experimental studies ceased to take place (Alam et al., 2021). These unique circumstances showed the need for updating the research behaviour to ensure continuous research even when direct contact between the researcher and the participants is not possible. In the field of psycholinguistics, even after the height of the pandemic, most studies are still conducted in laboratories and require direct contact with the participants. However, for some of the methods applied, there might be an alternative (Groot & Hagoort, 2018). With the growing digitalisation, the development of psycholinguistic online software and the improving technical abilities of the participants some experimental paradigms could be used outside of the laboratory. One possibility is designing experiments that can be completed online without temporal or spatial restrictions.

Furthermore, being able to conduct studies online offers a great solution to represent a broader range of the speaker population than laboratory studies normally do. Since a large proportion of participants in laboratory studies are students, the results of those studies might not be representative for all speakers of a language (Groot & Hagoort, 2018). In online studies it may be possible to include varying age groups and professional groups since living in proximity to an university or laboratory is not a necessity for a participation in a study. Admittedly, it is possible that a majority of the participants may still be students but the opportunity to recruit participants from different (cultural or social) backgrounds outside the university sphere may be higher than in laboratory studies.

Another important factor is the wider reach of online studies which means that more participants may be recruited and the backgrounds of the participants may be diverse, making the results of a study more universal. Speakers with physical disabilities or elderly speakers that are not able to go into a laboratory can be included and participate in online studies. Moreover, speakers of minority languages or languages counting only a few speakers can be reached more easily, helping the investigation of linguistic phenomena in these languages. All

these factors highlight the benefits that the expansion of psycholinguistic methods to the web might bring.

One of the main problems with online studies is the lack of control of the environment in which the study is taken. Although most instructions explicitly mention that the study should be taken in a quiet environment without any distractions or disturbances, the adherence to these measures cannot be monitored. Since the measured effects in psycholinguistic studies such as reaction times (RTs) can be very small and the variance in the environment of the participants might influence those effects, the uncontrolled environment presents one of the main reasons why the appliance of online experiments in the field of psycholinguistics is scarce (Groot & Hagoort, 2018). An in-depth comparison of the results of a psycholinguistic study conducted in a laboratory and an identical study conducted over the web might help to determine whether the uncontrolled environment has an influence on the results.

The aim of this Bachelor thesis is to investigate whether an online study can replicate the results of a laboratory study. It therefore examines the possible influence that the uncontrolled environment outside of laboratories and the possible variance in participants might have. The masked morphological priming paradigm in second-language processing for derived and inflected English words specifically will be investigated by trying to replicate the results of a laboratory study done by Ciaccio & Veríssimo (2020) in a web-based study.

2. Theoretical background

Priming is a phenomenon in which the processing of a word with a particular form is facilitated by the preceding word with a similar or related form (Pickering, 1999). The word that appears on a screen for a designated amount of time and is followed by a second word is called the prime. The second word is usually displayed longer than the prime and is called the target word. The prime word may be related to the target word in some way (e.g. orthographically) and may facilitate the processing of the target word. This facilitation can be measured with the help of reaction times. In this context, reaction time refers to the amount of time that passes between perceiving a word and responding to it (e.g. by pressing a button) (Fang & Davis, 2017). Morphological priming is a form of priming where the prime is morphologically related to the target word. An example for this is when the word *printed* is preceded by the word *print* (written as printed: PRINT from here onward). In this paradigm the prime is typically a morphologically complex word. A morphologically complex word is made up of a root (the stem of the word) and one or more morphemes that are added to the stem (e.g. the ending *–ed*) (Larsen & Nippold, 2007). In morphological priming the reaction times of morphologically related primes and target words are measured against a baseline condition. In the baseline condition the target words are preceded by morphologically unrelated primes such as cleaned: PRINT. The baseline condition is used as an indicator for normal reaction times when perceiving a certain word and reacting to it. The difference between the reaction times of the morphologically related condition and the unrelated (baseline) condition is referred to as priming effect (Drews, 1996). An example of an occurring priming effect is when the reaction times of the morphologically related condition are shorter than the reaction times of the baseline condition (Sonnenstuhl et al., 1999). In a regular morphological priming study the primes are presented long enough to be perceived consciously. On the contrary, in a masked morphological priming study the primes will not be seen consciously because they are presented very briefly and therefore are beneath the conscious perception threshold. But since priming effects appear, it can be assumed that the primes are perceived subconsciously (Grainger et al., 1991).

Words that may be used as primes in a masked morphological priming paradigm (and will be used in this study) are derived and inflected words. Derived and inflected words add different types of information to the stem of a word. Derivation changes the category of a word, e.g. $\text{print}_{\text{verb}} \rightarrow \text{printer}_{\text{noun}}$ and can add semantic information to a word (e.g. printer = a device that

prints), whereas inflection adds grammatical information to a stem (as required by the syntactic structure), for example tense ($\text{print}_{\text{present}} \rightarrow \text{printed}_{\text{past}}$) and does not add any semantic information. This results in the fact that derivation seems to produce new lexemes (units that carry a meaning) while inflection only yields new word forms of an already existing lexeme. In languages such as English or German both, derivation as well as inflection, are realised by affixation (Jacob et al., 2018).

A typical morphological masked priming study consists of the participants seeing a morphologically complex word form as a prime (e.g. *printed*) for a very brief amount of time, typically around 30 to 70 milliseconds (ms), and seeing either a morphologically related or unrelated word immediately afterwards as a target word (e.g. *print* or *clean*). The primes are presented within a visual mask (see Appendix) to avoid the conscious perception of the word. In a masked priming study the participants normally have to perform a lexical decision task on the target word (Jacob et al., 2018). In a lexical decision task the participants have to decide whether a presented stimulus is an existing word or not and react accordingly, e.g. by pressing a button (Meyer & Schvaneveldt, 1971).

Morphological priming effects are a robust finding from previous research on morphological processing. The occurrence of a morphological priming effect can be seen when the responses to the target word are facilitated (e.g. faster reaction times) following a morphologically related prime such as printed: PRINT compared to when a target word is preceded by an unrelated prime such as cleaned: PRINT (Feldman & Soltano, 1999). For the emergence of this speed-up effect three different interpretations have been proposed.

One proposal is the mechanism of morphological decomposition by stripping off affixes (Rastle et al., 2004). According to this account, the affixes are stripped from their stems pre-lexically (before the word is retrieved from the mental lexicon) during the masked priming task and therefore the stem gets pre-activated. This means that for the word *printed* the affix -*ed* will be removed from the stem *print* before the word is retrieved from the mental lexicon. This process ensures that the stem *print* gets activated without the affix hindering the activation and the prime *printed* will be accessed as *print* + *ed*. That results in the fact that following the affix stripping, only the stem *print* will be searched in the mental lexicon, causing faster reaction times when encountering the stem as a target word later (Rastle et al., 2004). These faster reaction times are the result of two different stages of morphological processing: the morpho-orthographic decomposition (orthographically based decomposing of

words in morphemes simply because they have a morphological structure) takes place very early in the stage of word processing, whereas the subsequent morpho-semantic analysis occurs at a later point (Rastle & Davis, 2008).

The second account assumes that during processing stems are activated as embedded words without a detailed morphological representation of the stem. During processing the stems at the edge of words dominate over other possibly embedded words due to the fact that the stem is typically the longest embedded word and the outer-letter processing advantage. The outer-letter processing advantage is an effect in which the outer letters of a string (at the beginning and at the end of a string) have a processing advantage over the other letters of a string. Therefore, the segmentation of the stem of a morphologically complex word from its affixes is facilitated if thestem is at the outer edge of a string, e.g. *print* from *print-ed* (Grainger & Beyersmann, 2017).

Alternatively, the third account proposes that the morphological priming effects are based on the shared semantic and orthographic properties of the prime and the target. It assumes that the faster reaction times are the result of a more or less semantically transparent meaning between the prime and the target word, therefore assuming no morphological representation during processing (Baayen et al., 2011). However, Beyersmann et al. (2016) refuted this hypothesis by proving that semantically opaque prime words such as cashew: CASH result in faster reaction times as well, thus making it highly unlikely that only semantically transparent primes cause a priming effect.

Previous studies using the masked morphological priming in second language processing have focused on the effect of derived and inflected words on the reaction times of a target word. The results differ from each other but almost all studies found significant priming effects for derived words (Jacob et al., 2018; Silva & Clahsen, 2008) whereas only some studies found a significant priming effect for inflected words (Coughlin & Tremblay, 2015; Voga et al., 2014). Although there are some differences between the results of those studies, it is evident that robust priming effects can be measured in non-native processing. To the best of my knowledge, all these masked morphological priming studies took place in a laboratory and the paradigm of masked morphological priming in non-native speakers has not been investigated online.

This Bachelor thesis examines whether an online study using the masked morphological priming paradigm can replicate the results obtained in a laboratory study and therefore help us

draw conclusion about the way in which the masked morphological priming paradigm can also be investigated online. To my knowledge, there is no existing study that has tried to replicate the results of a laboratory study using the masked morphological priming paradigm in non-native speakers in a web-based study.

3. Hypotheses

The laboratory study that will be used as a reference was conducted by Ciaccio and Veríssimo (2020). The study examined the variability in performance levels of non-native speakers in a masked morphological priming paradigm while measuring reaction times to targets preceded by either derived, inflected or unrelated primes. This online study will solely focus on the reaction times of the participants evoked by the primes and their corresponding target word. It will not take the variability in performance levels into account. Ciaccio and Veríssimo (2020) found that derived as well as inflected primes sped up the word recognition to an almost similar extent. This indicates existing priming effects for both types of morphological primes.

This online experiment tries to replicate the results of the laboratory study that were found by Ciaccio and Veríssimo (2020). It is expected to find a difference between the reaction times when a target word is preceded by a morphologically unrelated prime versus when it is preceded by a derived or inflected prime. Ideally, the reaction times for the derived and inflected condition should be very similar to one another. Distinct priming effects should be observed for the derived and inflected primed condition when compared with the baseline (unrelated) condition.

The expected outcome is that the results of the online study and the laboratory study resemble one another so that conclusions about the suitability of online studies for the masked morphological priming paradigm can be drawn.

4. Study design

4.1 Materials

The materials used in this online experiment were identical to those used in the laboratory study by Ciaccio and Veríssimo (2020) to minimise the risk of confounds. They created three different lists with 102 English monomorphemic verbs as targets. These targets were either preceded by an inflected prime (the –ed past tense form), a derived prime (the nominalised verb) or an unrelated prime. Half of the unrelated primes were –ed inflected forms and the other half were –er derived words. Ciaccio and Veríssimo (2020) tried to keep the word-form frequency, the lemma frequency and the length of the word as similar as possible in the different prime types. They used the SUBTLEX-UK database (van Heuven et al., 2014) to extract the word-form and lemma frequency. This database was created using the words and word-pairs of subtitles of nine British television channels to obtain a frequency score for English words. Van Heuven et al. (2014) created the Zipf scale which expresses the frequency values of words with values ranging from 1 to 7. Low frequency words are indicated by values below 3 whereas high frequency words are indicated by values above 4.

The targets and their primes were distributed across three lists using a Latin Square Design. This means that each participant saw 34 targets that were associated with each of the three prime types. Thus, each participant saw 34 target words preceded by an unrelated prime (e.g. cleaned: PRINT), 34 target words preceded by a derived prime (e.g. printer: PRINT) and 34 target words preceded by an inflected prime (e.g. printed: PRINT). Each target word was only seen once by each participant, ensuring that the target word does not elicit a priming effect due to repeated appearance. By reversing the order of the three lists, Ciaccio and Veríssimo (2020) created three additional lists. This resulted in six presentation lists in total.

Apart from the experimental pairs each lists contained 438 fillers, adding up to a total of 540 items per list. The primes for the filler were all existing words whereas 270 of the filler targets were non-existing English words created with the software Wuggy (Keuleers & Brysbaert, 2010), thus requiring a “no” response from the participants. Of the 438 filler pairs, 51 non-existing word targets were preceded by an inflected prime and another 51 by a derived prime. Therefore, neither inflection nor derivation could be used as a clue for the lexicality of the targets. To ensure that orthographical overlap does not prompt a “yes” response, 68 of the non-existing filler targets were orthographically embedded in their primes. Considering these factors, 25.2% of all prime-target pairs in each list were either morphologically or

orthographically related (Ciaccio & Veríssimo, 2020) (see Appendix for presentation lists and materials).

Additionally, all participants took the LexTale test (Lemhöfer & Broersma, 2012) which is a standardised vocabulary test. It consists of a lexical decision task without a time limit. This test was integrated at the end of the online experiment to ensure an adequate English level of the participants. The LexTale score is calculated by measuring the percentage of correct responses and correcting it for the proportion of existing English words and non-existing words in the test. A percentage between 80-100% indicates an C1 or C2 level, a percentage between 60-80% shows a B2 level and percentages below 60% indicate an English level of B1 or lower (Lemhöfer & Broersma, 2012).

4.2 Brief summary of the experimental procedure in Ciaccio and Veríssimo (2020)

The experimental procedure of the laboratory study that is aimed to be replicated will be explained briefly to facilitate the comparison between the online study and Ciaccio and Veríssimo's (2020) study.

The authors tested their participants in a quiet laboratory room; the testing session was carried out in English. The participants were evenly assigned to one out of six lists. The reaction times and the accuracy rates were measured with the help of the experimental software DMDX (Forster & Forster, 2003). The participants were informed about the procedure of the experiment and could ask questions if there still was any uncertainty about the task. They had to perform a lexical decision task on English words or non-words. The answers were indicated by pressing one of two buttons on a gamepad. The display of the primes and targets and the design of the masks were identical to those used in the online experiment (for a more detailed description, see procedure of the online experiment below).

The participant group of Ciaccio and Veríssimo (2020) included sixty-nine advanced non-native English speakers with a language level corresponding to a B2 to C2 level according to the LexTale test (Lemhöfer & Broersma, 2012). Their mean age was 26.09 years. They were all native speakers of German and started learning English after the age of 4 (mean age of acquisition = 8.71 years, standard deviation (SD) = 1.95 years, range of years = 4–13; 3 NAs (not available)).

4.3 Procedure of the online experiment

The experiment was designed as an online experiment with the help of the web-based version of DMDX (Forster & Forster, 2003). DMDX is a Windows based program used in psycholinguistic studies about language processing and is "designed to precisely time the presentation of text ... to enable the measurement of reaction times to these displays with millisecond accuracy." (Forster & Forster, 2003, p. 116). Further development of DMDX has made it possible to deploy DMDX experiments over the Internet. This updated version of DMDX is called webDMDX and does not require a communication over the web during the experimental session. The web version of DMDX can be executed by combining a self-extracting .zip file, a rich text file (.rtf) that specifies the parameters and commands used for the experiment (such as display duration of the primes or targets and the buttons that have to be pressed) and the input (e.g. the experimental lists), a poster.exe, a Windows .exe file that sends the participants' data to the researcher and a .bat file that contains the commands necessary to run the experiment (Witzel et al., 2015).

In this online experiment the webDMDX version was used to eliminate the possible confound of stability and speed of the participants' Internet connection. If the input for the experiment and the response data are communicated over the web between the testing computer and the participants' computer during the experimental session, the speed of the participants' Internet connection could influence the display and response timing. To avoid this confound, experiments running with webDMDX are run locally on the computer of the participant and the output will be transmitted over the web once the experiment is completed (Witzel et al., 2015).

The experiment was packed in a self-installing .exe file that started automatically after the participants downloaded it. The .exe file was created using the Winzip Software (https://www.winzip.com/en/). The .exe file consisted of a start.bat file, a sslposter.exe file and the dmdx.exe file from Forster (2021) as well as one of the item lists with the parameters, commands and input needed for the experiment and the LexTale test. The parameters and commands used in the .rtf file were as similar as possible to those used in Ciaccio and Veríssimo (2020) to avoid possible confounds. The only adjustment that needed to be made was the modification of the device name that the participants used for the experiment and the buttons that are pressed since Ciaccio and Veríssimo (2020) used a gamepad whereas the participants of the online experiment used a keyboard. Moreover, the video mode had to be modified to fit the online format of the study. The lists used as an input in the .rtf file were

identical. The accuracy and reaction times were measured by the experimental software webDMDX (Forster & Forster, 2003).

The communication of the instructions and the distribution of the participants between the six lists was enacted via the creation of a Google form (Google Ireland Limited). Google form was used since it is a well-known and familiar tool for most people. The participants were assigned one of the experimental lists by selecting one out of six numbers in a Google form. After choosing a number the participants received a link to a Dropbox (Dropbox International Unlimited Company, 2007) where they could download the experiment. The Cloud Storage Software Dropbox was used since it is widely known, easy to handle and accessible for all types of users. The most important reason for choosing Dropbox was that the user does not need to have an account for downloading the .exe file.

The instructions in the Google form instructed the participants on how to download the experiment and about the two parts of the experiment. Furthermore, the form included a declaration of consent that the participants had to agree to in order to start the experiment. If they did not agree, they were redirected to the end of the form without being able to participate in the experiment (see Appendix for instructions).

In the experiment, it was explained that the participants would see a series of English words and they had to decide whether the word is an existing word or an invented word. They had to indicate their answer as quickly and as accurately as possible. This task was a lexical decision task. The “yes” or “no” responses were provided by pressing a button on the keyboard.

The “J”-button had to be pressed when indicating that it is an existing word (a “yes” response, e.g. “glue”), the “N”-button had to be pressed when indicating that it is an invented word (a “no” response, e.g.”plun”). The “yes” response was provided with the dominant hand of the participant. The reason for choosing the “J” and “N” button on the keyboard was to include right-handed people as well as left-handed people since both buttons are grouped closely together so that the participant can press either of them with the right or left hand.

The procedure and the design of the experiment were kept as close as possible to the procedure and design used in Ciaccio and Veríssimo (2020).

For each list ten training trials were included at the beginning to ensure that the task was understood correctly. Each participant saw 540 targets during the experiment. In the course of

the experiment the participants had two breaks whose length they could choose freely. They could continue the experiment by pressing the button “B”.

At the beginning of each trial a blank screen was presented for 500 ms, followed by a mask consisting of ten hashes presented for 500 ms. Following the mask, the prime appeared for 50 ms and directly after, the target appeared and remained on the screen until a button was pressed or the predetermined timeout of 2000 ms was reached (see Appendix). The next trial began automatically after a button press or a timeout without giving feedback to the participant. The primes were always presented in lowercase letters whereas the targets were presented in uppercase letters (e.g. printed: PRINT).

The LexTale test (Lemhöfer & Broersma, 2012) was included in the experiment to avoid having participants with an insufficient level of English. It was bound to appear after the participants finished their chosen list with all 540 items. The LexTale test had to be completed after the experimental part was finished and appeared automatically. After running the experiment, the reaction times and accuracy responses for the online experiment and the Lextale test of each participant were separately sent to the researcher by email.

4.4 Participants

The participants were recruited using the Participant Pool of the Cognitive Sciences of the University of Potsdam (SONA) and a circle of friends. Thirty participants (16 women; 15 men) took part in the experiment in exchange for course credits. All participants had normal hearing, normal or corrected-to-normal vision, and no record of neurological diseases. Their mean age was 25.29 years with a range from 18 to 38 years and a standard derivation of 5.04 years. All participants were native speakers of German, none spoke another native language.

They demonstrated an English level ranging from intermediate to advanced, all started learning English after the age of 5 (mean age of acquisition = 8.51 years, SD= 1.95 years, range= 5 to 12 years, 2 NAs) and lived in Germany at the time of testing. Their skill in English was tested by means of the LexTale test (Lemhöfer & Broersma, 2012). The participants mean score was 77.93% (corresponding to a good B2 level) with a standard derivation of 10.54% within a range from 58.75% to 96.25%, corresponding to a B1 to C2 level. No participant had to be excluded due to insufficient English skills.

5. Challenges

Several challenges had to be overcome to ensure that the online study replicated the laboratory study as accurately as possible.

First of all, the correct display of the experiment within the visual mask had to be guaranteed to avoid that this confounding factor affects the comparison of the experiments. To solve this problem and achieve an optimal display quality on different laptops and computers, the video mode feature of DMDX was set to <vm desktop>. This configuration ensured that the trial was presented in a manner that was suited to the resolution and refresh rate of the screen to avoid different manners of displaying the experiment.

Another concern was the variability of the keyboard sampling rates due to a magnitude of existing hardwares. However, given the fact that this issue might arise even between different computers within one laboratory, it was deemed to be irrelevant. For the sake of certainty, the study of Enochson and Culbertson (2015) was consulted which investigated the variability problems of keyboard sampling rates and monitor refresh rates. The study showed that the magnitude of hardware differences is not broad enough to impact the results significantly.

Furthermore, it was assumed to be difficult to ensure an equal distribution of the participants across the different lists. This problem was solved by creating a Google form in which the participants had to choose one out of six numbers in order to get access to one of the six links leading to the experiment. During the course of the study the experimental links could be interchanged and replaced by a link leading to another list. This measure was implemented to control the distribution of the participants across the six lists and proved to be effective. To avoid personal preference (e.g. preferring even numbers, preferring numbers that can be divided by a specific number...) influencing the selection of numbers and possibly affecting the distribution of participants across the lists, all numbers available for selection were prime numbers.

One of the biggest challenges in designing online studies is the construction of the instructions for the participants. In a laboratory study the participants can ask the researcher for clarification if they do not understand the task, in an online study this is not possible (Groot & Hagoort, 2018). A solution for this was the careful composition of the instructions describing every small step that the participants might have to do (see Appendix). To ensure

the easy comprehensibility of the instructions and that every small step has been explained, the instructions of the online experiment were test read by several unbiased persons that do not have a background in linguistics. This offers a high degree of certainty that the instructions were easily understood by the participants and did not leave possible questions unanswered. Despite this test, it remained a potential risk that individual participants did not understand the instructions correctly.

A further concern was ensuring the relative comparability of the group of subjects participating in the laboratory study and the online study. This was solved by using the Prescreen Restrictions in the SONA participant pool. The same prescreen restrictions that were applied by Ciaccio and Veríssimo (2020) were applied to the online study, namely specifying German as native language, ensuring that the participants did not have neurological, hearing or visual impairments and asking for further clarification whether other language skills besides German and English exist. Moreover, an age restriction from 18 to 39 years was implemented to exclude older or younger participants. Those measures facilitated the process of having relatively similar groups of subjects in the online and the laboratory study.

One challenge that remained unsolved was the problem of the Windows virus protection. After the participants downloaded the .exe file and tried to open it, the virus protection popped up informing the participants that they tried to open a self-installing file from an unknown user. The participants had to disable the virus protection in order to run the experiment which was explained in the instructions in the Google form. Nevertheless, this appearance of the virus protection window could not be prevented since the researcher is not a certified software developer for Microsoft Windows.

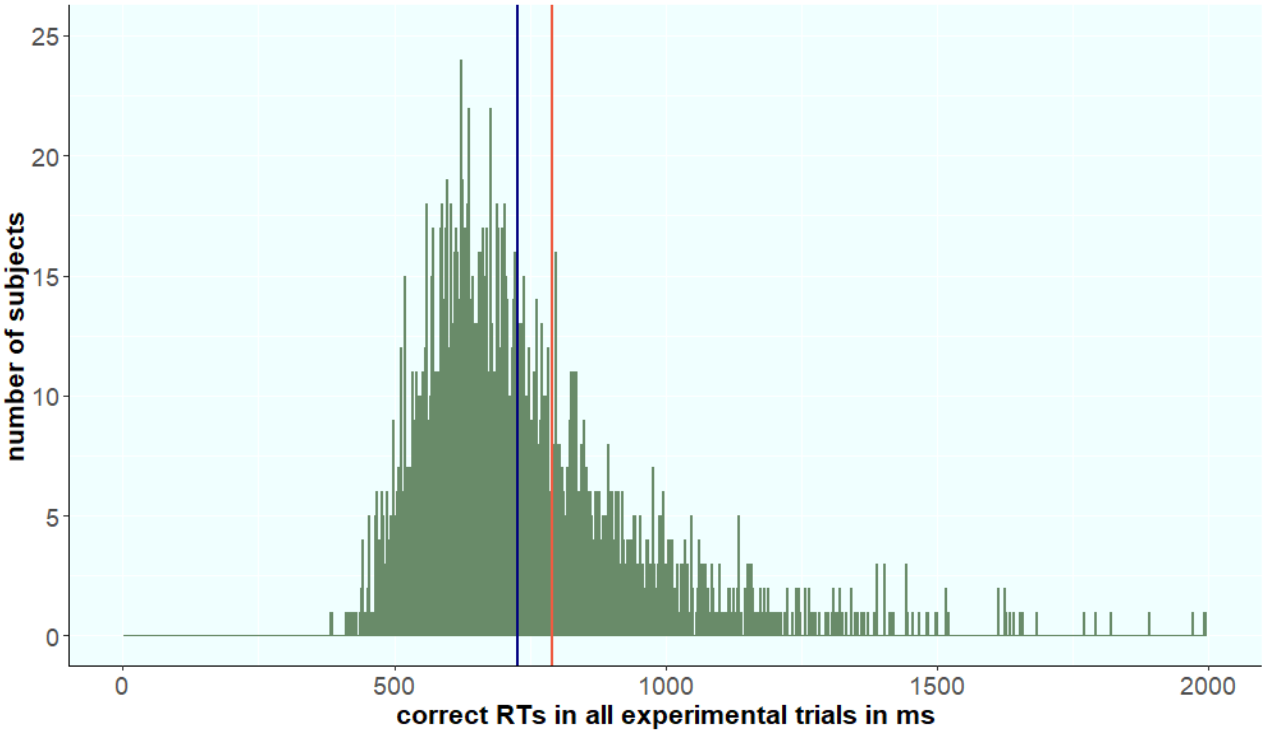
6. Results of the online study

All data analysis was carried out with the assistance of the statistical computing programme R (R Core Team, 2017) (see Appendix for analysis scripts and raw data). All subjects had high to medium accuracy rates in the word conditions. In the word trials the accuracy rates consisted of a range from 63.7% to 98.51% whereas in the non-word trials the accuracy rates showed a wider span ranging from 23.7% to 96.67%. An exclusion criterion was the below chance performance in non-word or word trials. This means that participants performing lower than 50% in their mean accuracy ratings for either word or non-word trials will be excluded from further analysis. Since it is highly probable that participants which scored below chance level are not able to discriminate between English words or non-words, their reaction times might be unreliable for a priming study. Four participants were excluded from analysis due to below chance performance in accuracy rating of the non-word trials. No items were excluded. Incorrect responses (12.53%) and timeouts (0.92%) were excluded from further analysis as well.

Reaction times were analysed using the statistical computing programme R (R Core Team, 2017). After visual inspection of the distribution of the reaction times, reaction times over 1.700ms and below 200ms were considered outliers and removed from the data and further analysis (1.42% of all data points, see Fig. 1). The minimum threshold of 200ms was chosen because the minimum reaction time for college-age individuals is around 200ms making faster reaction times a very unlikely response to the stimuli. The upper threshold of 1700 ms was chosen since a substantial gap exists between the mean reaction times after 1500 ms to 1700 ms (see Fig. 1). This indicates that the reaction times after 1700 ms can be seen as anomalous. As can be surmised from the Figure 1, the distribution of the mean reaction times values is rising rapidly on the left and has a long tail on the right. This results in the median appearing earlier than the mean in the data distribution (see Fig. 1), making it a right-skewed distribution. This is a typical distribution for reaction time values and can be interpreted by using three parameters: the mean, the standard deviation and the median (Whelan, 2008).

Figure 1

Reaction times of all experimental trials with correct responses (outliers included) together with the median (red line) and the mean (blue line)



The mean is used since it has a high significance for reaction time values although it might be greatly influenced by outliers (Bortz & Schuster, 2010). The mean reaction times for the three prime type conditions amount to 728 ms for the derived condition, 734 ms for the inflected condition and 775 ms in the unrelated condition (see Fig. 2). The means of the reaction times were shorter following the derived and inflected primes (by 47ms for derived primes and 41 ms for inflected primes). The means of the log-reaction times (to simulate a normal distribution) for inflected and derived primes are also smaller than the mean of the log-reaction times for the unrelated primes. The log-values estimates are mentioned for the sake of completeness. The difference in reaction times for both, derived and inflected primes, compared to the unrelated primes seems to indicate a facilitation effect for the lexical decision task.

The median for all reaction times included in the analysis was 790 ms. For the derived condition the median was 683 ms, for the inflected condition the median was slightly higher with 692 ms (see table 1). The unrelated condition had the highest median amounting to 727 ms. The median was included as an additional control measure. Baayen and Milin (2010) recommend using the median for non-symmetrical distributions such as reaction times

because the median is less sensitive than the mean to the skew of the distribution. With the help of the median it is possible to indicate a central tendency of the distribution of the data points. For all three conditions, the median reaction times are faster than the mean reaction times by roughly 6%, indicating that most reaction times are faster than the mean. This suggests a wide range of the data points despite the removing of outliers.

The standard deviation shows a decrease in variability in the derived and inflected condition compared to the unrelated condition. The standard deviation for the unrelated condition is 211 ms while the SDs for the derived condition (196 ms) and for the inflected condition (190 ms) show a more compressed picture, being smaller by 15 ms and 21 ms respectively (see table 1). This indicates that the data points are more clustered around the mean and there is less variability in the reaction times for both conditions.

Table 1

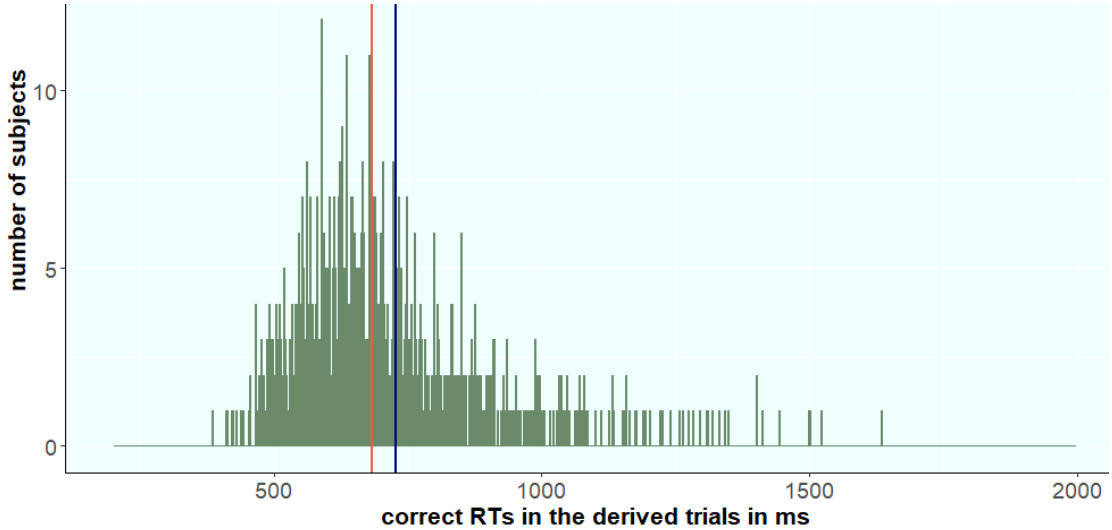
Summary of the median reaction times, mean reaction times and standard deviations for all three experimental conditions with their priming effect

Prime type	Median of the reaction times in ms	Mean reaction times in ms	Standard deviation in ms	Priming effect in ms
unrelated	726.88	775	211	-
Derived	683.46	728	196	47
inflected	692.01	734	190	41

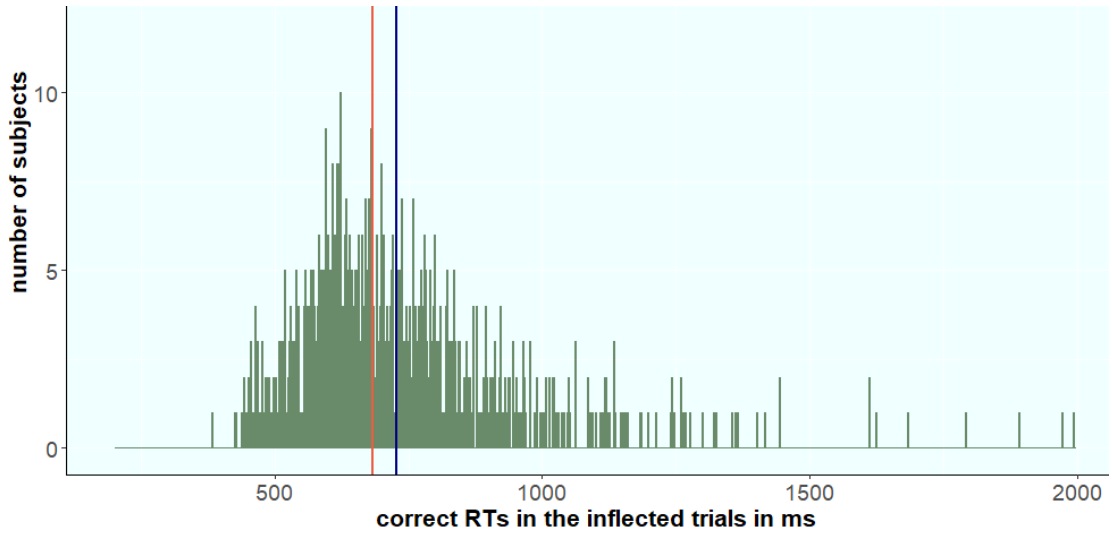
Figure 2

Reaction times of the trials with correct responses (outliers included) together with the median (red line) and the mean (blue line)

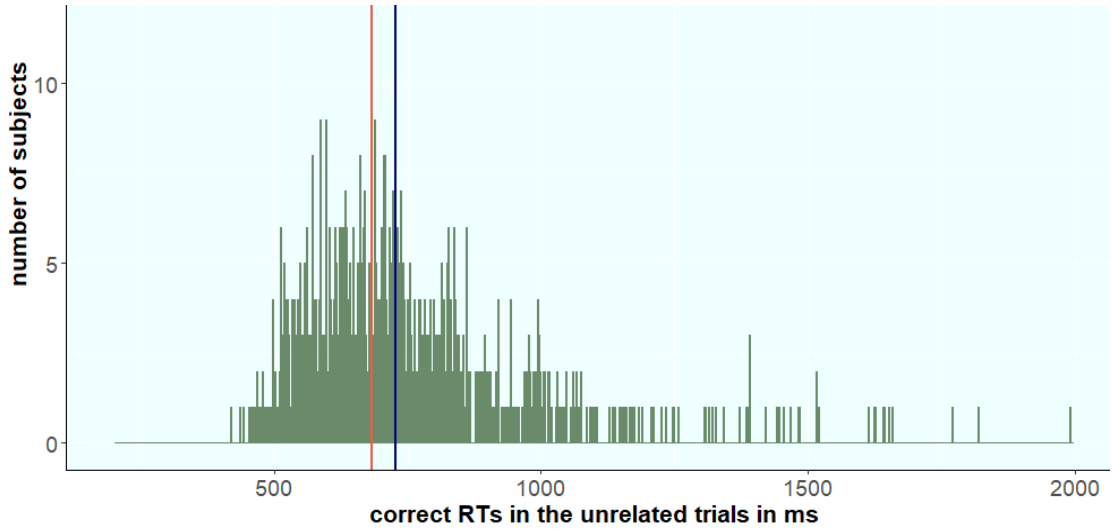
(a) derived condition



(b) inflected condition



(c) unrelated condition



7. Discussion

7.1 Prior findings for the masked morphological priming paradigm

The present study investigated the masked morphological priming effects on the mean reaction times to target words that are preceded by either a derived prime, inflected prime or an unrelated prime in an online study. Previous studies have shown that morphological priming in a non-native second-language (L2) speakers is a robust finding for derived primes but absent or reduced for inflected primes.

Silva and Clahsen (2008) conducted a masked-priming study comparing derivational and inflectional primes in English to investigate whether both, native speakers (L1) and L2 speakers, display priming effects for both types of primes. The L2 speakers showed a partial priming effect for derived forms but no priming effect for inflected forms. A similar effect was found by Kirkici and Clahsen (2013) as a result of a masked-priming study investigating Turkish derivational and inflectional forms in L1 and L2 speakers. Similarly, the L2 group showed priming effects for the derivational condition but not for the inflectional condition. A study directly comparing masked-priming effects in German by Jacob et al. (2018) confirms the results mentioned above. Again, L2 speakers showed a significant priming effect for derivational primes but no facilitation for inflectional primes.

However, several priming studies found significant morphological priming effects in L2 speakers for inflected forms. Voga et al. (2014) conducted a masked priming study examining the effect of derivational and inflectional primes in Greek L2-speakers of English. In this study, the L2 group demonstrated a robust and significant priming effect for both, derivational and inflectional forms. Another masked-priming study by Coughlin and Tremblay (2015) with L2 speakers of French found a significant morphological priming effect for inflectional forms, albeit the study did not include any derivational primes. Feldman et al. (2010) investigated the morphological priming effect for inflectional primes with L2 speakers of English with Serbian as their native language. The results show further evidence for the fact that there is an existing significant priming effect for inflectional forms.

The results of Ciaccio and Veríssimo's (2020) study align with the aforementioned studies insofar that a significant priming effect for inflectional as well as derivational primes can be observed to an almost similar extent.

The results of this online study show a priming effect for derived as well as inflected forms, mirroring the results of Ciaccio and Veríssimo (2020). However, in Ciaccio and Veríssimo (2020) the mean reaction times for inflected forms were slightly faster than for derived forms whereas in this study the mean reaction times for derived forms were slightly faster than for inflected forms. Nevertheless, both studies show that the two types of primes speed up the word recognition process.

To offer a perspective on the question whether the results of masked morphological priming laboratory study can be replicated in an online study, an in-depth comparison of the results of the online study and the study done by Ciaccio and Veríssimo (2020) is needed.

7.2 Comparison of the results of the present online study and the laboratory study by Ciaccio and Veríssimo (2020)

Ciaccio and Veríssimo (2020) report mean reaction times of 672 ms for derived primes, 668 ms for inflected primes and 708 ms for unrelated primes.[1] The mean reaction times for the derived and inflected condition were faster than for the unrelated condition. The authors had a mean SD of 183 ms for the derived conditions, 180 ms for the inflected condition and 186 ms for the unrelated condition[1]. In the paper they reported a log-SD of 0.243 ms for the derived condition, 0.238 ms for the inflected condition and 0.245 ms for the unrelated condition.

In general, it can be said that the mean reaction times of the online study were longer. The mean reaction time in the derived condition for the online study was 56 ms slower than in the laboratory study. A similar picture emerges for the inflected condition, the difference amounting to 66 ms faster in Ciaccio & Veríssimo's (2020) results. Even in the unrelated condition a mean reaction time difference of 67 ms can be seen (see table 2).

The priming effect appears to be stronger in the online study, since the effect amounts to 47 ms for the derived condition whereas in the laboratory study the derived condition only showed a priming effect of 36 ms. This accounts for a difference of 11 ms between the two studies. However, in the inflected condition there exists almost no difference between the priming effects of the two studies. The online study reported a priming effect of 41 ms and the

[1] The mean reaction times in ms and the SDs in ms are not reported in the paper from Ciaccio and Veríssimo (2020) but in order to make a meaningful comparison, the non-transformed reaction time values were obtained from the authors.

laboratory study reported a priming effect of 40 ms. Notwithstanding the differences in the reaction times for the conditions, both types of studies found significant priming effects for the derived and inflected condition (see table 2).

Table 2

Summary of the mean reaction times for all three experimental conditions and their priming effect

Priming type	Mean in ms in the online study	Mean in ms in the laboratory study	Priming effect in ms in the online study	Priming effect in ms in the laboratory study
Unrelated	775	708	-	-
Derived	728	672	47	36
Inflected	734	668	41	40

The median reaction times in the online study were slower across all three conditions compared to the laboratory study. Ciaccio and Veríssimo (2020) report median reaction times of 661 ms for the unrelated condition and 625 ms for the inflected and derived condition (see table 3). Although the median reaction times in the online study were slower, there is still a robust difference between the reaction times of the unrelated condition and the derived and inflected condition insofar that the derived and inflected conditions show faster reaction times. This difference can also be seen in the median values of Ciaccio and Veríssimo (2020). The difference between the unrelated condition and the inflected and derived condition respectively, is 36 ms. In the online study this difference amounts to 44 ms for the derived condition and 35 ms in the inflected condition. It can easily be seen that both studies report a higher median for the unrelated condition, indicating the existence of a priming effect. In the inflected condition the median reaction times of the laboratory and the online study are almost identical in their variation from the median reaction times of the unrelated condition. In the derived condition the median reaction time of the online study deviates stronger from the unrelated condition median reaction time than the laboratory study. The difference between those two studies amounts to 8 ms, suggesting a stronger facilitation effect for the derived condition in the online study. But since this difference is still relatively small, it does not indicate a meaningful difference between the two studies.

Table 3

Summary of the median reaction times for all three experimental conditions in the online and laboratory study

Priming type	Median in ms in the online study	Median in ms in the laboratory study
Unrelated	726.88	661
Derived	683.46	625
Inflected	692.01	625

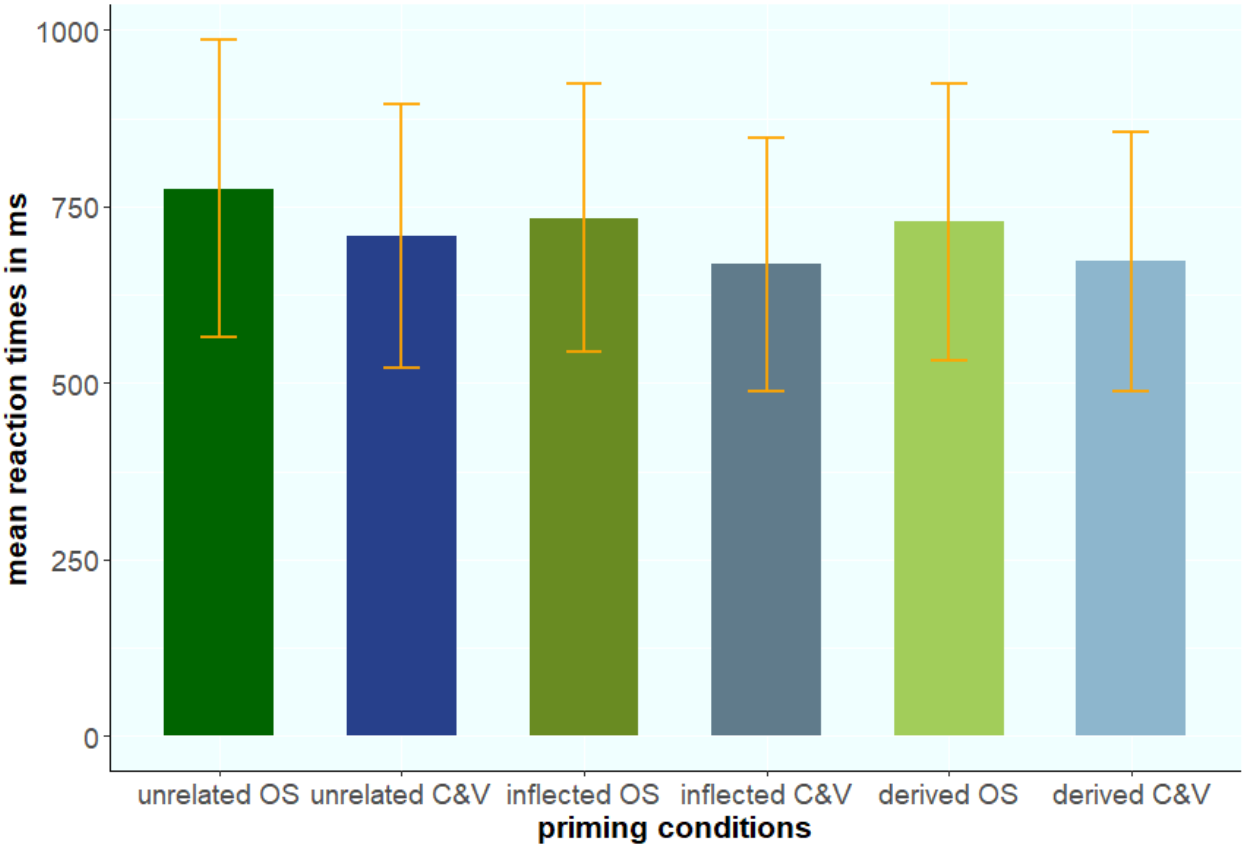
The standard deviation for derived primes was reported as being 184 ms in the derived condition, 180 ms in the inflected condition and 186 ms in the unrelated condition (Ciaccio & Veríssimo, 2020). When comparing these values to the values of the SD in the online study, it is clearly visible that the SD values in the online study are bigger. The biggest difference can be seen in the values of the SD in the unrelated condition, with the difference being 25 ms bigger than in the laboratory study. The difference in the SDs for the inflected condition is the smallest (10 ms) whereas the difference for the derived condition is slightly bigger with 12 ms. The difference between the SDs of the two studies shows that the reaction times in the online study are generally more dispersed than in the laboratory study. The participants of the online study exhibit a greater variance regarding their reaction times, especially in the unrelated condition. The derived and inflected conditions are almost similar in how they differ from the SD of the laboratory study.

Although the standard deviations of the online study are bigger, they are still in line with the standard deviations that can be seen in previous laboratory masked priming morphological priming studies with non-native speakers. Kirkici and Clahsen (2013) and Jacob et al. (2018) stated ranges of standard deviations that were only slightly smaller than those found in the online study and as mentioned above, the standard deviations reported in Ciaccio and Veríssimo (2020) were only slightly smaller as well. Therefore, the bigger standard deviations of the online study do not present a robust difference to the laboratory study. To some extent a difference between the standard deviations of an online study and a laboratory study can be expected due to the distractions that might take place during web-based studies (see Limitations below) and increase the variability. Considering the standard deviations of the online study and therefore the variability of the reaction times are well within a normal range

of standard deviations for masked morphological priming studies, it can be assumed that the priming effects which were found in the online study are reliable and robust.

Figure 3

Comparison of the mean reaction times in the laboratory and online study



Note. This figure demonstrates the mean reaction times of the three experimental conditions in the laboratory study by Ciaccio and Veríssimo (2020) (C&V) and the online study (OS); each bar also contains the SD (orange line). The green bars represent the three different conditions of the online study and the blue bars represent the three different conditions of the laboratory study.

In figure 3 it is clearly visible that both studies found a significant priming effect for derived as well as inflected primes. Although the reaction times in the online study for the derived condition were slightly faster than for the inflected condition, there is no reason to assume that there is a reliable difference between the two conditions. A priming effect with a difference of 6 ms between the two conditions does not indicate a robust numerical difference

and is neglectable, making the two conditions comparable in their priming effect. The same applies to the priming effects in Ciaccio and Veríssimo (2020). Albeit the priming effect for inflected words is faster by 4 ms, the numerical difference of those values is so small that it can be neglected.

The longer reaction times in the online study point to the possibility of a main effect of group. As can be seen in the mean reaction times (see Fig. 3), one group of participants (the group in the laboratory experiment) tended to be faster than the other group of participants (the group in the online study) across all conditions. The higher standard deviation in the derived and inflected conditions suggests a higher variance. This means the slower reaction times of the participant group in the online study leave more room for variability, possibly influencing the results and the priming effect.

Wagenmakers and Brown (2007) reported a linear relationship between the mean and the standard deviation such that larger reaction times lead to larger standard deviations (and therefore larger variability). Considering this finding, the higher standard deviation in the online study compared to the laboratory study is not surprising since the online study exhibits longer reaction times in all three conditions. The observed priming effects in the online study are slightly bigger than the priming effects reported in Ciaccio and Veríssimo (2020) which might be due to the greater variability that is present in the online study.

But despite the greater variance in the reaction times of the online study and the generally longer reaction times, the online study found a robust priming effect for derived as well as inflected primes when compared with the baseline (unrelated) condition. A priming effect of 47 ms (derived condition) and 41 ms (inflected condition) respectively can be seen as a reliable difference from the baseline condition even with the presence of larger standard deviations. The study by Ciaccio and Veríssimo (2020) found significant priming effects for both priming conditions as well.

Taken together, the mean reaction times and the median reaction times show the overall longer reaction times in all three conditions in the online study. However, the priming effects of the derived and inflected primes are clearly evident and point to a facilitation of the lexical decision. These results align with the results from the laboratory study which found a significant priming effect for both, derived and inflected primes and a resulting facilitation effect.

Considering these findings, the results of the laboratory study and the online study can be interpreted as comparable. This might be seen as evidence for the fact that online studies are suitable for the investigation of the masked morphological priming paradigm in non-native speakers.

7.3 Limitations

The results of the online study highlight several difficulties that may appear when trying to replicate a laboratory study with the masked morphological priming paradigm over the web. These difficulties comprise the uncontrolled experimental environment, the independence of the participants, the high variation of the participants' backgrounds, the group size in the online study, and the technical difficulties when conducting an online study. These limitations may have some influence on the results and will be discussed below.

First, one possible reason for the longer reaction times in the online study could be the environment in which the study was carried out. Since it was an online study that could be taken at anytime and anywhere, the researcher had no control about the study environment and could not eliminate potential confounds. The participants could have taken the study at night or early in the morning when they were not fully concentrated. One third of the participants of the online study took the experiment later than 7 pm. The participants that decided to take the experiment at night might already have strained their eyes during screen activities during the day which might lead to longer reaction times. A laboratory study is normally conducted throughout the day, making it highly unlikely that the participants are as exhausted as they might have been during the online study. Besides, there is the possibility of disturbances and disruptions such as children or loud external sounds. It might be the case that the online format of the study contributed to the lack of concentration since the participants might not have taken the study as seriously as they would have done it when taken in the laboratory. These factors may have influenced the results and therefore contributed to the longer reaction times in the online study. In Ciaccio and Veríssimo's (2020) study the environment could be controlled by the researchers and possible confounds or a lack of concentration could have been detected and avoided.

A further problem was posed by the independence of the participants. The participants can take the experiment at any time, therefore for most online experiments of this kind, the researcher cannot be reached if questions arise. This might lead to confusion on the part of the participants, resulting in errors that could be prevented in an on-site laboratory setting. The inaccessibility of the researcher might have contributed to the fact that four participants had to be excluded from the analysis due to below chance performance. Perhaps this below chance performance might have been avoidable in the laboratory where the participants could have asked questions to clarify the task.

Another reason for the longer reaction times of the participants and the higher variance in the reaction times in the online study might be the background of the participants. In the online study almost 14 participants (47%) were non-students. In the study by Ciaccio and Veríssimo (2020) only 9 participants (13%) were non-students.[2] Since only half of the participants in the online study had a university background, it might be viable to assume that the longer reaction times could result from a different usage of English of the non-student participants. Even if the level of English was being determined through the means of the LexTale test, there were major differences between the levels of the participants, the English level ranging from B1 to C2 with a SD of 10.54%. In Ciaccio and Veríssimo (2020) the English level of the participants was more homogenous with the English levels only ranging from B2 to C2 and a SD of only 9.28%. Furthermore, the participants without a university background might not be used to participating in this kind of studies, resulting in longer reaction times. Although this might present a valid reason for the longer reaction times, it cannot be ruled out that some of the non-student participants might have been students previously.

In order to compensate for the lack of concentration, the possibility of confounds and the variability of the participants' background it is advisable to include a greater number of participants in online studies than would be the case in a laboratory study. Additionally, the need for more trials in an online study may arise since more trials would help to counterbalance the possible confounds and influence of noise on the data and variations.

This brings up a further problem since the number of participants in the online study was less than half of the number of participants in the laboratory study which might have lead to the longer reaction times. Less participants means that the influence of each participants' reaction times on the mean and median had a bigger impact than it would be the case in a big group of participants. The results of the online study might have been differently with more participants. One reason for the low participation rate could be the relatively short recruitment period for the online experiment which has been only 5 weeks and that the participants only received course credits for their participation. However, more participants were enrolled than actually participated in the experiment. Many of those enrolled participants might have started the experiment and never completed it for different reasons. This may have lead to a higher drop-out-rate than in laboratory studies since the drop-out-rate in laboratory studies is almost

[2] The number of students and non-students is not mentioned in the paper by Ciaccio and Veríssimo (2020). The numbers were obtained from the authors to offer a perspective for a comparison of the composition of the group of subjects.

0% (Mathot & March, 2021). Since the researcher only received the results of the participants that completed the experiment, it cannot be determined how many participants started the experiment but never finished it. Moreover, the participants might not have felt as obligated to complete the experiment as they would have felt in the laboratory especially since the rules of the Participant Pool of the Cognitive Sciences of the University of Potsdam do not entail consequences in the case of non-participation in online experiments in contrast to when a participant is enrolled in a laboratory study and does not participate. Another reason for the higher drop-out-rate might be the virus protection warning since this could have discouraged some of the enrolled participants to download the experiment and complete it. It would be advisable to try and solve the problem with the virus protection before conducting further online studies.

Another factor that might have influenced the reaction times is the refresh rate of the screens of the participants. Since the webDMDX-code used in this experiment measured the display duration of the primes and target words in ticks as indicated by <fd 30> and the duration of a tick depends on the refresh rate of the screen, it might be possible that the computers used by the participants have different refresh rates (Garaizar et al., 2014). This might result in slightly different representations of the primes and targets on the screen. It would have been better to change the frame duration parameter in the webDMDX-code to <%ms 50> to ensure that regardless of the screen refresh rate all participants see the primes and target for the same amount of time (50 ms).

One obstacle that was not foreseen was that four participants did not take the LexTale test since they closed the experiment after finishing the experimental part. Due to the processing time of the reaction times of the experimental part caused by the separate transmission of the reaction times of the experiment and the LexTale test to the researcher, the change from the experimental part to the LexTale test could take up to one minute. Although it was explicitly mentioned in the instructions in the Google form that the experiment consists of two parts and that a change will occur after the first part, some participants did not wait until the appearance of the LexTale test. Since for those four participants the English level could not be determined it is possible that their English level was below B1. But the mean reaction times of these participants did not show a noticeable difference from the reaction times of the other participants hence their reaction times were not excluded. For future online studies it might be advisable to avoid sending the reaction times of the experimental part to the researcher before changing to the LexTale part.

8. Conclusion and Outlook

The present study investigated whether the results obtained in a masked morphological priming laboratory study with non-native speakers (Ciaccio & Veríssimo, 2020) can be replicated in an online study. It looked at the facilitation effects that derived or inflected primes have on their target words. Both studies were conducted with the psycholinguistic software DMDX (Forster & Forster, 2003). The results of the laboratory and the online study were compared with the help of the statistical values mean, median and standard deviation. Both, the laboratory and the online study, found evidence for the fact that target words preceded by derived or inflected primes produce faster reaction times than when preceded by unrelated words. Although the online study showed longer reaction times in general, the priming effect elicited by the derived or inflected primes was comparable to the priming effect of the laboratory study. Both studies found that derived and inflected primes affect the reaction times to an almost similar extent making the two primes comparable in their facilitation effect. Both types of primes facilitate the processing of their targets whereas unrelated primes do not affect processing times. It is evident that the laboratory study and the online study yielded comparable results for the priming effects in the derived and inflected condition.

The results show that an online study using the masked morphological priming paradigm can replicate the results obtained in a laboratory study. Thus, the masked morphological priming paradigm with non-native speakers can be investigated over the web and the application of online studies in the masked morphological priming paradigm is a viable alternative to laboratory studies. Even if it is important to recognise the limitations of online studies, the present study indicates that the uncontrolled environment and variance in participants does not present a problem even for relatively small-scaled reaction time values. With the appropriate adjustment of the experimental procedure web-based studies might present a viable alternative to laboratory studies even for highly time-sensitive experiments such as masked morphological priming. This opens up the possibility for the investigation of other psycholinguistic paradigms that do not require to be conducted in a laboratory over the web.

With the help of online studies it is possible to reach a greater number of people who may be willing to participate in a psycholinguistic study. As a consequence of this higher number of participants, the group of participants might have a greater variability owing to the different social, cultural and linguistic backgrounds making the group of subjects less homogenous.

Furthermore, the participants can theoretically participate from all around the world, offering the possibility of having participants from different regions of a country or different linguistic communities. It is easier to reach isolated communities or to investigate languages with only a few speakers when using online studies.

Although, web-based studies seem to present a feasible alternative to laboratory studies for the masked morphological priming paradigm with non-native speakers, it remains open to debate whether this conclusion can also be applied to studies with native speakers. Furthermore, since the present study only investigated the masked morphological priming paradigm it remains to be seen whether online studies are a viable alternative for other psycholinguistic methods or paradigms. In order to investigate this, further research needs to be carried out.

9. References

Alam, A., Rampes, S. & Ma, D. (2021). The Impact of the COVID-19 Pandemic on Research. *Translational Perioperative and Pain Medicine*, *8*(1). https://www.researchgate.net/publication/348309913_The_Impact_of_the_COVID-19_Pandemic_on_Research

Baayen, R.H. & Milin, P. (2010). Analyzing Reaction Times. *International Journal of Psychological Research*, *3*(2), 12–28. https://dialnet.unirioja.es/servlet/articulo?codigo=3405162

Baayen, R. H., Milin, P., Đurđević, D. F., Hendrix, P., & Marelli, M. (2011). An amorphous model for morphological processing in visual comprehension based on naive discriminative learning. *Psychological Review*, *118*(3), 438–481. https://doi.org/10.1037/a0023851

Beyersmann, E., Ziegler, J. C., Castles, A., Coltheart, M., Kezilas, Y., & Grainger, J. (2016). Morpho-orthographic segmentation without semantics. *Psychonomic Bulletin & Review*, *23*(2), 533–539. https://doi.org/10.3758/s13423-015-0927-z

Bortz, J., & Schuster, C. (2010). *Statistik für Human- und Sozialwissenschaftler* (7., vollständig überarbeitete und erweiterte Auflage). *Springer-Lehrbuch*. Springer.

Ciaccio, L. A., & Veríssimo, J. (2020). *Investigating variability in morphological processing with Bayesian distributional models.* https://doi.org/10.31234/osf.io/qk5cu

Coughlin, C. E., & Tremblay, A. (2015). Morphological decomposition in native and non-native French speakers. *Bilingualism: Language and Cognition*, *18*(3), 524–542. https://doi.org/10.1017/s1366728914000200

Drews, E. (1996). Morphological Priming. *Language and Cognitive Processes*, *11*(6), 629–634. https://doi.org/10.1080/016909696387033

Dropbox International Unlimited Company. (2007). *Dropbox*. https://www.dropbox.com/de/

Enochson, K., & Culbertson, J. (2015). Collecting psycholinguistic response time data using Amazon mechanical Turk. *PLOS ONE*, *10*(3), e0116946. https://doi.org/10.1371/journal.pone.0116946

Fang, J. Y., & Davis, T. L. (2017). Reaction Time in Parkinson's Disease☆. In J. Stein (Ed.), *Reference module in neuroscience and biobehavioral psychology.* Elsevier. https://doi.org/10.1016/B978-0-12-809324-5.00770-7

Feldman, L. B , & Soltano, E. G. (1999). Morphological priming: The role of prime duration, semantic transparency, and affix position. *Brain and Language*, *68*(1-2), 33–39. https://doi.org/10.1006/brln.1999.2077

Feldman, L. B, Kostić, A., Basnight-Brown, D. M., Durđević, D. F., & Pastizzo, M. J. (2010). Morphological facilitation for regular and irregular verb formations in native and non-native speakers: Little evidence for two distinct mechanisms. *1366-7289*, *13*(2), 119–135. https://doi.org/10.1017/S1366728909990459

Forster, J. C. (2021, August 18). *Remote Testing Overview*. http://psy1.psych.arizona.edu/~jforster/dmdx/help/dmdxhremotetestingoverview.htm

Forster, K. I., & Forster, J. C. (2003). Dmdx: A windows display program with millisecond accuracy. *Behavior Research Methods, Instruments, & Computers : A Journal of the Psychonomic Society, Inc*, *35*(1), 116–124. https://doi.org/10.3758/BF03195503

Garaizar, P., Vadillo, M. A., López-de-Ipiña, D., & Matute, H. (2014). Measuring software timing errors in the presentation of visual stimuli in cognitive neuroscience experiments. *PLOS ONE*, *9*(1), e85108. https://doi.org/10.1371/journal.pone.0085108

Google Ireland Limited. *Google forms*. https://www.google.com/forms/about/

Grainger, J., & Beyersmann, E. (2017). Chapter Nine - Edge-Aligned Embedded Word Activation Initiates Morpho-orthographic Segmentation. In B. H. Ross (Ed.), *Psychology of Learning and Motivation* (Vol. 67, pp. 285–317). Academic Press. https://doi.org/10.1016/bs.plm.2017.03.009

Grainger, J., Colé, P., & Segui, J. (1991). Masked morphological priming in visual word recognition. *Journal of Memory and Language*, *30*(3), 370–384. https://doi.org/10.1016/0749-596X(91)90042-I

Groot, A. M. B. de, & Hagoort, P. (Eds.). (2018). *Guides to research methods in language and linguistics: Vol. 9. Research methods in psycholinguistics and the neurobiology of language: A practical guide*. Wiley Blackwell.

Jacob, G., Heyer, V., & Veríssimo, J. (2018). Aiming at the same target: A masked priming study directly comparing derivation and inflection in the second language. *International Journal of Bilingualism*, *22*(6), 619–637. https://doi.org/10.1177/1367006916688333

Keuleers, E., & Brysbaert, M. (2010). Wuggy: A multilingual pseudoword generator. *Behavior Research Methods*, *42*(3), 627–633. https://doi.org/10.3758/BRM.42.3.627

Kirkici, B., & Clahsen, H. (2013). Inflection and derivation in native and non-native language processing: Masked priming experiments on Turkish. *Bilingualism: Language and Cognition*, *16*(4), 776–791. https://doi.org/10.1017/s1366728912000648

Larsen, J. A., & Nippold, M. A. (2007). Morphological Analysis in School-Age Children: Dynamic Assessment of a Word Learning Strategy. *Language Speech and Hearing Services in Schools*, *38*(3), 201–212. https://doi.org/10.1044/0161-1461(2007/021)

Lemhöfer, K., & Broersma, M. (2012). Introducing LexTALE: A quick and valid Lexical Test for Advanced Learners of English. *Behavior Research Methods*, *44*(2), 325–343. https://doi.org/10.3758/s13428-011-0146-0

Leminen, A. (2012). Neurocognitive processing of inflected and derived words. https://helda.helsinki.fi/handle/10138/28798

Mathot, S., & March, J. (2021). *Conducting linguistic experiments online with OpenSesame and OSWeb*. https://www.researchgate.net/publication/349185659_Conducting_linguistic_experiments_online_with_OpenSesame_and_OSWeb https://doi.org/10.31234/osf.io/wnryc

Meyer, D. E., & Schvaneveldt, R. W. (1971). Facilitation in recognizing pairs of words: Evidence of a dependence between retrieval operations. *Journal of Experimental Psychology*, *90*(2), 227–234. https://doi.org/10.1037/h0031564

Pickering, M. (1999). Syntactic priming in language production. *Trends in Cognitive Sciences*, *3*(4), 136–141. https://doi.org/10.1016/s1364-6613(99)01293-0

R Core Team. (2017). *R: A language and environment for statistical computing*. URL https://www.R-project.org/

Rastle, K., & Davis, M. H. (2008). Morphological decomposition based on the analysis of orthography. *Language and Cognitive Processes*, *23*(7-8), 942–971. https://doi.org/10.1080/01690960802069730

Rastle, K., Davis, M. H., & New, B. (2004). The broth in my brother's brothel: Morpho-orthographic segmentation in visual word recognition. *Psychonomic Bulletin & Review*, *11*(6), 1090–1098. https://doi.org/10.3758/BF03196742

Silva, R., & Clahsen, H. (2008). Morphologically complex words in L1 and L2 processing: Evidence from masked priming experiments in English. *Bilingualism: Language and Cognition*, *11*(2), 245–260. https://doi.org/10.1017/S1366728908003404

Sonnenstuhl, I., Eisenbeiss, S., & Clahsen, H. (1999). Morphological priming in the German mental lexicon. *Cognition*, *72*(3), 203–236. https://doi.org/10.1016/S0010-0277(99)00033-5

van Heuven, W. J. B., Mandera, P., Keuleers, E., & Brysbaert, M. (2014). Subtlex-UK: A new and improved word frequency database for British English. *Quarterly Journal of Experimental Psychology (2006)*, *67*(6), 1176–1190. https://doi.org/10.1080/17470218.2013.850521

Voga, M., Anastassiadis-Symeonidis A., & Giraudo H. (2014). Does morphology play a role in L2 processing? Two masked priming experiments with Greek speakers of ESL. *Lingvisticæ Investigationes*, *37*(2), 338–352. https://www.researchgate.net/publication/275958916_Voga_Anastassiadis-Symeonidis_Giraudo_2014_Does_morphology_play_a_role_in_L2_processing_Two_masked_priming_experiments_with_Greek_speakers_of_ESL

Wagenmakers, E.-J., & Brown, S. (2007). On the linear relation between the mean and the standard deviation of a response time distribution. *Psychological Review*, *114*(3), 830–841. https://doi.org/10.1037/0033-295X.114.3.830

Witzel, J., Cornelius, S., Witzel, N., Forster, K.I. & Forster, J.C. (2015). Testing the viability of webDMDX for masked priming experiments. In *Phonological and phonetic considerations of lexical processing* (pp. 169–198). https://doi.org/10.1075/bct.80.07wit

Whelan, R. (2008). Effective Analysis of Reaction Time Data. *The Psychological Record*, *58*(3), 475–482. https://doi.org/10.1007/BF03395630

Title page

ZIM – Zentrum für Informationstechnologie und Medienmanagment (n.d.). *Logos der Universität Potsdam*. Universität Potsdam. Retrieved 13.11.2021, from https://www.uni-potsdam.de/fileadmin/projects/zim/images/logos/04_Humjpg.jpg

10. List of figures and tables

Figure 1: *Reaction times of all experimental trials with correct responses* 18
Figure 2: *Reaction times of the trials with correct responses* 20
Figure 3: *Comparison of the mean reaction times in the laboratory and online study* 25
Figure A1: *The presentation of the prime and target in the visual masks* 38

Table 1: *Summary of the median reaction times, mean reaction times and standard deviations* 19
Table 2: *Summary of the mean reaction times for all three experimental conditions and their priming effect* 23
Table 3: *Summary of the median reaction times for all three experimental conditions in the online and laboratory study* 24

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Statement of Originality

I hereby declare that this Bachelor thesis is the product of my own work.

All the assistance received in preparing this thesis and the sources used have been acknowledged. All passages which I have taken from other sources are marked in the Reference list. I assure that this thesis has not previously been submitted for assessment either at the University of Potsdam or elsewhere.

Chiara Meisch 25.11.2021

Appendix

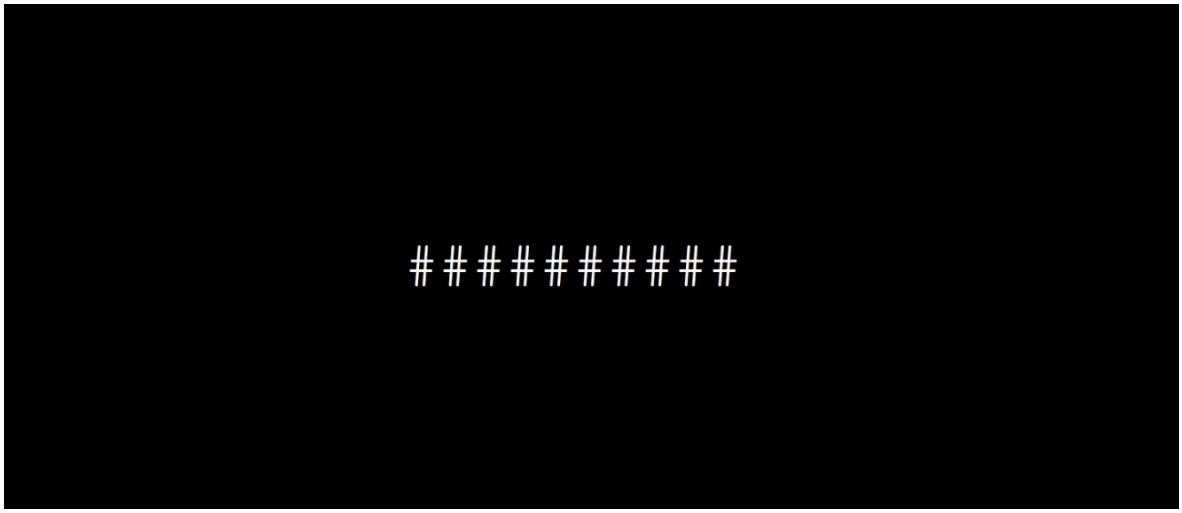
The instructions, the materials, the raw data, and the analysis scripts related to this Bachelor thesis are available at https://osf.io/2fra7/ .

Figure A1

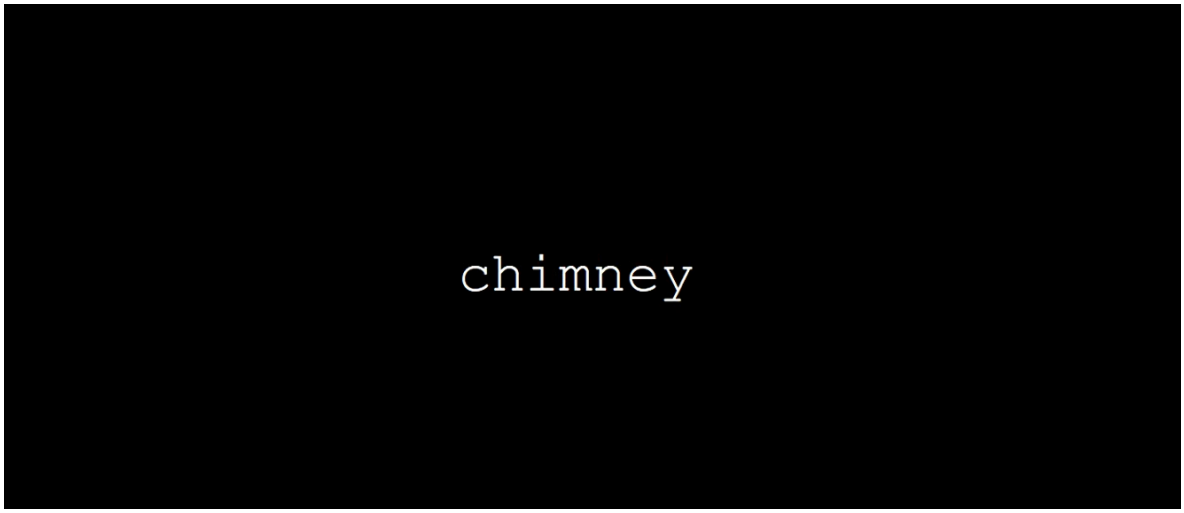
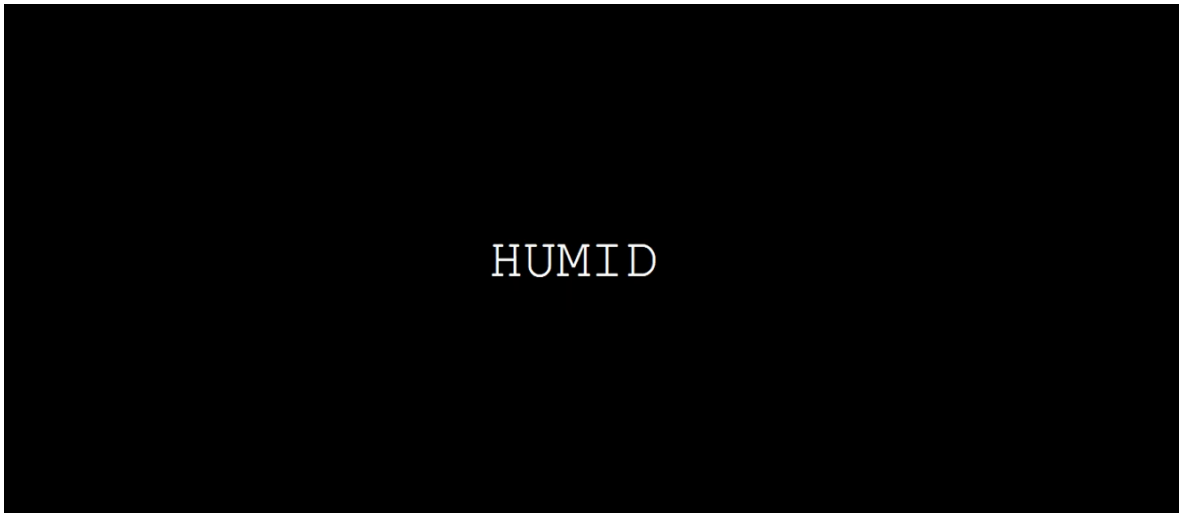
The presentation of the prime and target in the visual masks

(a) A blank screen: displayed for 500 ms

(b) A mask consisting of 10 hashes: displayed for 500 ms



(c) The (in this case unrelated) prime word: displayed for 50 ms



(d) The target word: displayed until a button press or the timeout at 2000 ms